

**UNIVERSITY OF THE FREE STATE
UNIVERSITEIT VAN DIE VRYSTAAT**

BLOEMFONTEIN CAMPUS/KAMPUS

CSIS3744

**DEPARTMENT: COMPUTER SCIENCE AND INFORMATICS
DEPARTEMENT: REKENAARWETENSKAP EN INFORMATIKA**

CONTACT NUMBER/KONTAKNOMMER: 051 401 2929

**EXAMINATION/EKSAMEN:
Main End-of-year/Hoofjaareind 2019
PAPER/VRAESTEL 1**

ASSESSOR(S)/

ASSESSOR(E): 1. Mnr. W.S.J. Marais

MODERATOR(S)/ 1. Mnr. C.A. Cilliers

MODERATOR(E): 2. Dr. B.P. Haskins

TIME/TYD: 180 min

MARKS/PUNTE: 100

INSTRUCTIONS: Answer all the questions. Number your questions clearly and correctly. The Afrikaans versions of the questions are in italics.

INSTRUKSIES: *Beantwoord al die vrae. Nommer u antwoorde duidelik en korrek. Die Afrikaanse weergawes van die vrae is kursief gedruk.*

Question / Vraag 1 [16 Marks / Punte]

Kovsie Attorneys implemented a 100Base-T Ethernet network with three public wireless access points and an Internet connection. Four administrative computers are connected to the wired network. Kovsie Attorneys doesn't employ any specialised IT personnel due to limited finances and therefore needs your help with evaluating their network. /

Kovsie Prokureurs het 'n 100Base-T Ethernet netwerk geïmplementeer met drie openbare draadlose toegangspunte en 'n Internet konneksie. Vier administratiewe rekenaars is aan die bedrade netwerk gekoppel. Kovsie Prokureurs het nie enige gespesialiseerde IT-personeel in diens geneem nie as gevolg van beperkte finansies en daarom benodig hulle jou hulp met die evaluering van hulle netwerk.

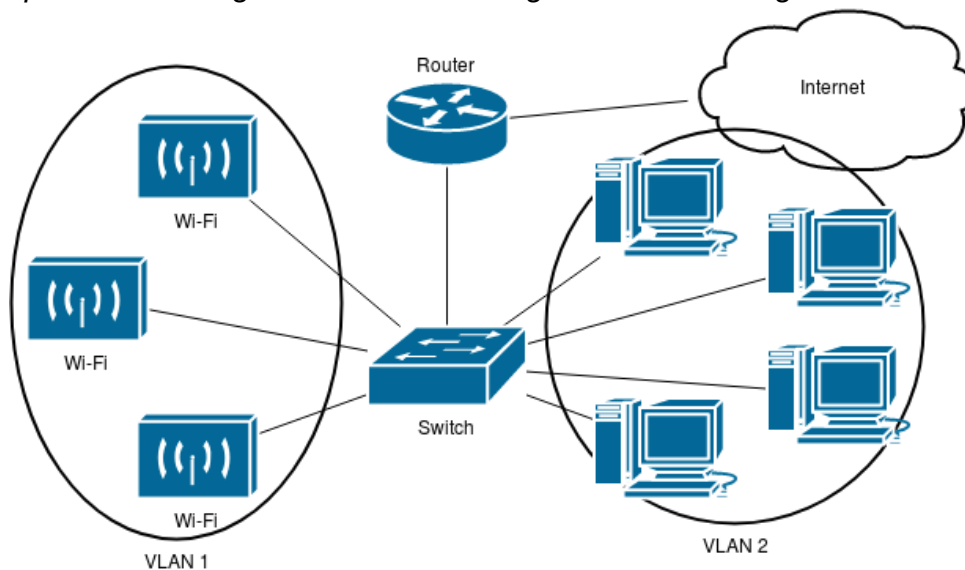
- 1.1 Explain the meaning of the Ethernet standard that has been implemented. /
Verduidelik die betekenis van die Ethernet standaard wat geïmplementeer is. (3)
- 1.2 Which physical media conforms to the Ethernet standard for this network? /
Watter fisiese media voldoen aan die Ethernet standaard vir hierdie netwerk? (1)
 - 1.2.1 Discuss three advantages with regard to this type of physical media. /
Bespreek drie voordele ten opsigte van hierdie tipe fisiese media. (3)

- 1.3 Give two reasons why Kovsie Attorneys implemented a peer-to-peer network rather than a client/server network. /
Gee twee redes waarom Kovsie Prokureurs eerder 'n eweknie-netwerk geïmplementeer het as 'n kliënt/bediener-netwerk. (2)
- 1.4 Explain to Kovsie Attorneys why it would be better to set up a client/server network by comparing the disadvantages of peer-to-peer networks with the advantages of client/server networks. /
Verduidelik aan Kovsie Prokureurs hoekom dit beter sal wees om 'n kliënt/bediener-netwerk op te stel deur die nadele van eweknie-netwerke met die voordele van kliënt/bediener-netwerke te vergelyk. (7)
 [16]

Question / Vraag 2 [24 Marks / Punte]

Kovsie Attorneys' current network address is 192.168.10.0. They want to separate their administrative computers from the wireless network by dividing the network into two VLANs as illustrated in the following network diagram: /

Kovsie Prokureurs se huidige netwerkadres is 192.168.10.0. Hulle wil die administratiewe rekenaars skei van die draadlose netwerk deur die netwerk in twee VLAN's op te deel soos geïllustreer in die volgende netwerkdiagram:



- 2.1 Give five reasons for using VLANs on a network. /
Gee vyf redes vir die gebruik van VLAN's op 'n netwerk. (5)
- 2.2 What type of connectivity device would the administrator have to reconfigure in order to set up the VLANs? /
Watter tipe verbindingstoestel sal die administrateur moet verstel ten einde die VLAN's op te stel? (1)
- 2.3 Explain how trunking protocols operate.
Verduidelik hoe stamprotokolle funksioneer. (2)
- 2.3.1 Why is Cisco's VLAN trunking protocol the most popular trunking protocol? /
Hoekom is Cisco se VLAN-stamprotokol die gewildste stamprotokol? (2)

- 2.4 Specify how many collision domains and how many broadcast domains Kovsie Attorneys' network will contain when the VLANs are implemented. /
Spesifiseer hoeveel botsing-domeine en hoeveel uitsaai-domeine Kovsie Prokureurs se netwerk sal bevat nadat die VLAN's geïmplementeer word. (2)
- 2.4.1 Motivate your answer by stipulating which portion of the network falls within the broadcast domain(s) and which portion falls within the collision domain(s). /
Motiveer jou antwoord deur te stipuleer watter deel van die netwerk binne die uitsaai-domein(e) val en watter deel in die botsing-domein(e) val. (2)
- 2.5 Kovsie Attorneys also need to assign each VLAN its own subnet of IP addresses by creating two subnets. /
Kovsie Prokureurs moet ook aan elke VLAN sy eie subnetwerk van IP-adresse toeken deur die skep van twee subnetwerke.
- 2.5.1 Calculate the new subnet mask for Kovsie Attorneys. /
Bereken die nuwe subnetmasker vir Kovsie Prokureurs. (5)
- 2.5.2 How many mobile devices will VLAN 1 be able to host? /
Hoeveel mobiele toestelle sal VLAN 1 in staat wees om te huisves? (1)
- 2.6 Complete the table below by supplying each subnet's network address, broadcast address and usable host address range: /
Voltooi die onderstaande tabel deur elke subnetwerk se netwerkadres, uitsaai-adres en bruikbare gasheer-adresreeks te verskaf:

	Subnet	Network Address <i>Netwerkadres</i>	Broadcast Address <i>Uitsaai-adres</i>	Usable Host Address Range <i>Bruikbare Gasheer-adresreeks</i>
2.6.1	1			
2.6.2	2			

(2)

(2)

[24]

Question / Vraag 3 [19 Marks / Punte]

Kovsie Attorneys' wireless access points are configured as follows: /
Kovsie Prokureurs se draadlose toegangspunte is as volg opgestel:

- Access point 1's SSID: Kovsie1WiFi / *Toegangspunt 1 se SSID: Kovsie 1WiFi*
- Access point 2's SSID: Kovsie2WiFi / *Toegangspunt 2 se SSID: Kovsie 2WiFi*
- Access point 3's SSID: Kovsie3WiFi / *Toegangspunt 3 se SSID: Kovsie 3WiFi*
- Encryption standard: WEP / *Enkripsiestandaard: WEP*
- DHCP: Enabled / *DHCP: Aangeskakel*
- Wi-Fi Protected Setup: Enabled / *WiFi Beskermd Opstelling: Aangeskakel*
- Supported 802.11 standards: n/ac / *Ondersteunde 802.11 standaarde: n/ac*

- 3.1 Identify and explain all the relevant security risks that are contained in the access points' configuration settings and give a solution for the vulnerabilities. /
Identifiseer en verduidelik al die tersaaklike veiligheidsrisiko's vervat in die toegangspunte se konfigurasie-instellings en gee 'n oplossing vir die kwesbaarhede. (4)

- 3.2 Name and discuss the protocol that is implemented by wireless access points to counter the “hidden node problem”. /
Noem en bespreek die protokol wat deur draadlose toegangspunte geïmplementeer word om die “verborge node problem” teen te werk. (4)
- 3.3 Draw a Wi-Fi Ethernet frame and label the different components of the frame. /
Teken 'n Wi-Fi Ethernet-raam en benoem die verskillende komponente van die raam. (4)
- 3.4 The throughput of the wireless LAN drops to 4 Mbps when there is more than 30 users. Describe the probable cause for the low throughput and explain how this pitfall can be avoided. /
Die deursetvermoë van die draadlose LAN daal na 4 Mbps wanneer daar meer as 30 gebruikers is. Beskryf die waarskynlike oorsaak vir die lae deursetvermoë en verduidelik hoe hierdie slagkat vermy kan word. (3)
- 3.5 Discuss the technologies, which allow 802.11n/ac to achieve such high throughput rates. /
Bespreek die tegnologieë, wat 802.11n/ac toelaat om so hoë deursetkoerse bereik. (4)
- [19]

Question / Vraag 4 [17 Marks / Punte]

Kovsie Attorneys has an ADSL Internet connection. /
Kovsie Prokureurs het 'n ADSL Internet konneksie.

- 4.1 Identify the router that connects Kovsie Attorneys' LAN to the Internet; is it an interior, exterior or border router? /
Identifiseer die roetebepaler wat Kovsie Prokureurs se LAN aan die Internet verbind; is dit 'n interieure-, eksterieure- of grensroetebepaler? (1)
- 4.1.1 On what level of the OSI model does a router operate? /
Op watter vlakke van die OSI-model funksioneer 'n roetebepaler? (1)
- 4.1.2 Discuss the OSPF link-state routing protocol. /
Bespreek die OSPF skakel-toestand roetebepalingsprotokol. (4)
- 4.2 What protocol is used for connecting home computers to an ISP via DSL? /
Watter protokol word gebruik vir die aansluiting van huisrekenaars aan 'n ISP via DSL? (1)
- 4.3 What server is necessary for users to connect to a LAN from a remote location using the Point-to-Point Protocol (PPP)? /
Watter bediener is nodig vir gebruikers om by 'n LAN aan te sluit van 'n afgeleë plek deur die Punt-na-Punt-Protokol (PPP) te gebruik? (1)
- 4.3.1 Why is Point-to-Point Protocol (PPP) a better protocol than Serial Line Internet Protocol (SLIP)? /
Waarom is Punt-tot-Punt Protokol (PPP) 'n beter protokol as Serie-Lyn Internet Protokol (SLIP)? (5)

- 4.4 Discuss the difference between a site-to-site VPN and a client-to-site VPN. /
Bespreek die verskil tussen 'n "site-to-site" VPN en 'n "client-to-site" VPN. (4)
 [17]

Question / Vraag 5 [12 Marks / Punte]

The DNS server, dns05.server.net, has the following IP address: 146.182.15.25. Its IP address was assigned by the DHCP server. /
Die DNS-bediener, dns05.server.net, het die volgende IP-adres: 146.182.15.25. Sy IP-adres is deur die DHCP-bediener toegeken.

- 5.1 Discuss the type of addressing that is implemented on the Transport, Network and Datalink layers of the OSI model respectively. /
Bespreek die tipe adressering wat onderskeidelik op die Transport-, Netwerk- en Datalink-vlakke van die OSI-model geïmplementeer word. (3)
- 5.2 What console command can be used to determine whether the DNS server is responding? Write down the full command. /
Watter konsole-instruksie kan gebruik word om te bepaal of die DNS-bediener reageer? Skryf die volledige instruksie neer. (2)
- 5.2.1 Explain how this command uses ICMP and ARP to perform its task. /
Verduidelik hoe hierdie instruksie ICMP en ARP gebruik om sy taak uit te voer. (3)
- 5.3 What domain does the DNS server belong to? /
Aan watter domein behoort die DNS-bediener? (1)
- 5.4 Explain how the auto-configuration with regard to IPv6 addressing works. /
Verduidelik hoe die outo-opset ten opsigte van IPv6-adressering werk. (3)
 [12]

Question / Vraag 6 [12 Marks / Punte]

Study the following header and payload information of an intercepted network packet. The information has already been translated into readable text format. /
Bestudeer die volgende kopstuk- en "payload"-inligting van 'n onderskepte netwerkpakkie. Die inligting is reeds in leesbare teksformaat vertaal.

Bit 0-7		Bit 8-15			Bit 16-23			Bit 24-31		
4	20 bytes	0			40 bytes					
0						1	0	0		
60		TCP			Correct					
172.13.14.15										
192.168.1.1										
443					49153					
3043958669										
937013559										
20 bytes		0	1	0	0	1	0	3200 bytes		
Correct										

- 6.1 Which Application layer protocol was used during the transmission of the packet? /
Watter Toepassingsvlakprotokol is tydens die oordrag van die pakkie gebruik? (1)
- 6.2 What is the total size of the packet? /
Wat is die totale grootte van die pakkie? (1)
- 6.3 Provide the source and destination socket addresses for the transmission. /
Verskaf die bron- en bestemming-sokadrese vir die oordrag. (2)
- 6.4 How many hops can the packet still be forwarded before it will be discarded? /
Hoeveel "hops" kan die pakkie nog steeds aangestuur word voordat dit weggegooi word? (1)
- 6.5 Explain the purpose of the "3200 bytes" value in the header. /
Verduidelik die doel van die "3200 grepe" waarde in die kopstuk. (2)
- 6.6 The packet illustrated above was the second packet of a three-way handshake transmitted between a workstation and a webserver on the Internet. /
Die pakkie wat hierbo geïllustreer word, was die tweede pakkie van 'n drie-rigtinghanddruk wat tussen 'n werkstasie en 'n webbediener op die Internet oorgedra is.
- 6.6.1 Please illustrate how the third packet of the three-way handshake will look by specifying its sequence number, acknowledgment number, source socket, destination socket and transport layer flags that will be set. /
Illustreer asseblief hoe die derde pakkie van die drie-rigtinghanddruk sal lyk deur die volgnommer, erkenningsnommer, bron-sok, bestemming-sok en transportlaagvlaggies wat ingestel sal word, te spesifiseer. (5)
- [12]

End of Paper / Einde van Vraestel